

Trust-Flow-Driven Trusted Federated Learning with TEE-Anchored Admission in Edge Computing

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Abstract

Federated learning must distinguish an update produced by an admissible execution from an update that is useful and semantically safe under Non-IID data. A static trusted-execution gate cannot answer the second question, whereas a single update-quality threshold can remove legitimate long-tail clients. This paper presents Trust Flow, a system-level decision framework that keeps execution credibility, current update content, long-term utility, and short-term risk as separate evidence streams until aggregation. A TEE-anchored trusted management and attestation agent (TMAA) supplies admission evidence; an explicit Kalman recursion and a dual-stream Beta/EMA state model define cold-start and cross-round behavior; and a layer-wise rule performs risk-dependent clipping with data-size-aware survivor re-normalization. The evaluation uses two Intel/AMD TEE-capable hardware configurations, Flower/PyTorch, CIFAR-10, 20 admitted clients, six concurrent attack behaviors, five seeds per platform, and a 30% malicious setting. Targeted ASR is macro-averaged over trigger, clean-label, and semantic attacks, whereas untargeted attacks are evaluated through clean accuracy, loss, and state transitions. Under the stated protocol, Trust Flow reaches 92.31% clean accuracy, 10.21% targeted ASR, and no permanent false ban in the tested runs.

Keywords: Federated learning; Edge computing; Trusted execution environment; Risk state modeling; Layered robust aggregation; Backdoor defense

1 Introduction

Current advancements in terminal device computing power enable efficient local model training using on-device data. Meanwhile, to better aggregate multi-device training experiences for collaborative model optimization while preserving local data privacy, the federated learning paradigm has been extensively studied [1]. Its core feature—exchanging only model parameters without offloading data—is widely applied in distributed collaborative training, serving as a critical pillar for edge intelligence. However, as the interconnected network environment becomes increasingly open, the performance bottlenecks of federated learning no longer reside solely in model architecture or optimizer design. The trustworthiness and robustness of the aggregation phase have emerged as vital factors determining overall availability. In practical scenarios, edge devices often exhibit significant heterogeneity in both data distribution and processing capabilities. Furthermore, fully constraining the behaviors of participating clients remains challenging. Such characteristics inevitably amplify statistical drifts and introduce security risks during empirical aggregation.

In standard federated learning paradigms, clients upload parameters after completing local data processing and model training. The server then aggregates these local updates and distributes the updated global model, iterating this process until convergence. Evidently, this workflow encounters multiple vulnerabilities within the increasingly open, multi-edge interconnected network. Open training environments inherently lack mandatory integrity constraints. Malicious clients can bypass actual training, directly crafting and uploading poisoned parameters, thereby severely impairing the server’s ability to verify the authenticity of aggregated updates. Several studies propose methods based on trusted execution or verifiable training [2, 3, 4], employing hardware-assisted remote attestation to ensure local training authenticity. Nevertheless, these mechanisms fail to address scenarios where the training process is genuine but the model updates are injected with malicious features. Furthermore, attackers execute parameter poisoning or stealthy backdoor attacks [5, 6] by embedding malicious triggers within normal update distributions, drastically reducing the success rate of single-round detection. To alleviate the instability of single-round

assessments, researchers have introduced methods based on historical reputation and similarity [7, 8] that leverage cross-round information. Yet, these approaches suffer from delayed recognition of sleeper attacks, where adversaries accumulate reputation early on before engaging in persistent, low-amplitude poisoning. Additionally, single-reputation scores combined with fixed-threshold mechanisms remain highly inadequate for identifying deep abnormal trigger features.

The presence of non-independent and identically distributed (Non-IID) data across devices increases the directional dispersion of optimal updates. This heterogeneity easily blurs the boundary between normal statistical fluctuations and anomalous deviations, creating a conflict between robustness and generalization under strong Non-IID conditions. Certain approaches rely on distance-based exclusion or statistical truncation to suppress abnormal updates [9, 10], while others adaptively adjust weights to mitigate heterogeneous impacts [11]. Under severe long-tail distributions, however, such methods often misclassify legitimately drifted parameter updates as anomalies. This misjudgment can lead to the excessive exclusion of crucial heterogeneous information, significantly degrading the model’s generalization capability. In real-world edge intelligence scenarios, these challenges—open execution environments, stealthy cross-round attacks, and pronounced heterogeneous statistical drifts—rarely occur in isolation. Multiple factors frequently compound, leading to intractable dilemmas such as elevated false positive rates from tightening interception thresholds or increased false negative rates when relaxing them. Consequently, establishing a unified state modeling and coordinated decision-making mechanism spanning the entire workflow remains a critical challenge. Such a mechanism must balance client admission and trustworthiness during local training with low false positive and high recall requirements during aggregation, all while maintaining the convergence and generalization performance of collaborative model optimization.

To tackle the three major challenges outlined above, this paper proposes a trusted federated aggregation mechanism driven by a full-process trust flow, conceptualized from a hardware-software co-design perspective. This mechanism follows the execution pipeline of federated learning. During client admission and local training, TEE remote attestation, runtime feature entropy analysis, and Kalman filtering are introduced to formulate a dynamic admission trust score that updates across rounds. At the server auditing stage, the current-round content quality is evaluated based on a pure reference direction. The client’s state is then decoupled into a historical utility flow and an instant risk flow, utilizing Bayesian posterior inference and risk EMA to separately characterize long-term contributions and short-term anomalies. During the aggregation phase, layered risk gating, survivor weight re-normalization, and dynamic clipping are executed based on these trust states. This strategy suppresses malicious updates while retaining legitimate heterogeneous contributions, effectively mitigating the tradeoff between security and generalization under Non-IID conditions.

The primary contributions of this paper are system-level and concern the ordering and interface between established mechanisms rather than a new cryptographic primitive or Byzantine estimator:

1. **TEE-anchored admission with an explicit cold-start policy.** Remote attestation defines the execution boundary, while entropy-derived observations and a covariance-aware Kalman recursion produce a continuously updated admission score. A newly admitted client starts in NORMAL and cannot be permanently blacklisted in its first round.
2. **Dual-stream state evolution.** Explicit decayed Beta updates describe historical utility, while an EMA of the strongest current risk probe preserves short-term attack evidence. The two streams are fused only when aggregation privilege is computed.
3. **Risk-gated layered aggregation.** Layer sensitivity scores determine thresholds and clipping radii; the original data-volume weight is retained and only surviving mass is re-normalized.
4. **Protocol-matched evaluation.** Two Intel/AMD TEE-capable platforms, five seeds per platform, a shared 5,000-sample client-training pool plus a server-held 500-sample proxy set, a CIFAR-10-adapted ResNet-18, and six concurrent attack behaviors are evaluated. Recent defenses are discussed as positioning references and are not assigned unverified numerical results.

2 Background and Related Work

2.1 Federated Learning and Edge Computing Collaborative Architecture

Federated learning (FL) completes global model training through parameter-level collaboration under the constraint that data remain local. In edge computing scenarios, the client-edge-cloud architecture has been widely adopted: the edge side performs local training and uploads updates, while the central server handles aggregation and model distribution.

In the standard FL setting, the global optimization objective is to minimize the empirical risk over the data distributions of all clients:

$$\min_W \mathcal{L}(W) = \sum_{k=1}^K p_k \mathcal{L}_k(W) \quad (1)$$

where $p_k = |\mathcal{D}_k| / \sum |\mathcal{D}_j|$ denotes the data-volume proportion. In round t , the server distributes the global model $W_{global}^{(t-1)}$. Client k takes it as the initial model and runs SGD for E local epochs with learning rate η_l , yielding the local update:

$$\Delta W_k^{(t)} = W_{k,E}^{(t)} - W_{global}^{(t-1)}, \quad \text{where } W_{k,e}^{(t)} = W_{k,e-1}^{(t)} - \eta_l \nabla \mathbb{E}[\ell(W_{k,e-1}^{(t)}; x, y)] \quad (2)$$

Here, the expectation $\mathbb{E}[\cdot]$ denotes mini-batch sampling over the local dataset $\mathcal{D}_k$. The client then uploads its update, and the server applies a form of parameter aggregation: $W_{global}^{(t)} = W_{global}^{(t-1)} + \text{Agg}(\{\Delta W_k^{(t)}\}_{k \in \Phi})$.

This architecture, however, naturally faces communication latency, device heterogeneity, and Non-IID data. To mitigate these issues, prior work has proposed methods such as Clustered FL [12], adaptive pruning-expansion (FedPE) [13], and split federated learning (ParallelSFL) [14]. As system complexity increases, the attack surface also expands, giving malicious nodes more room to remain hidden and interfere with training.

2.2 Edge Impersonation, Model Poisoning, and Stealthy Backdoor Attacks

In open federated networks without mandatory admission control, the server has difficulty accurately judging client identities and update quality. Attacks mainly fall into two categories. The first is **untargeted poisoning**, such as Sign-flipping and Gradient Scaling, which aims to disrupt global convergence and cause denial of service. The second is **targeted backdoor** attack, including label flipping, clean-label poisoning, and semantic backdoors, which induce targeted misclassification at inference time through concealed triggers.

2.3 Mechanisms and Inherent Limitations of Existing Active Defenses

To address these threats, early defenses largely relied on geometric and statistical filtering, such as Krum[9] and Trimmed Mean[10]. More recent studies have introduced temporal trust evaluation, latent representations, and malicious-client filtering, including FLTrust[7], FoolsGold[8], RaSA[11], FLDetector[15], and FLARE[16], while continuing to expand into malicious detection[17], trusted training frameworks[2], hierarchical protection[18], and related directions. For backdoor and collusion attacks, schemes such as DeepSight[19], FLAME[20], CrowdGuard[21], FLPurifier[22], RoseAgg[23], and ShieldFL[24] have also been developed. Purely software-based defenses still have clear weaknesses under strong Non-IID conditions and high-privilege attacks. Tightening thresholds raises the false positive rate (FPR), while relaxing them may allow dormant attacks to pass. When attackers control the underlying environment, software probes that lack a hardware root of trust also struggle to form a complete defensive loop.

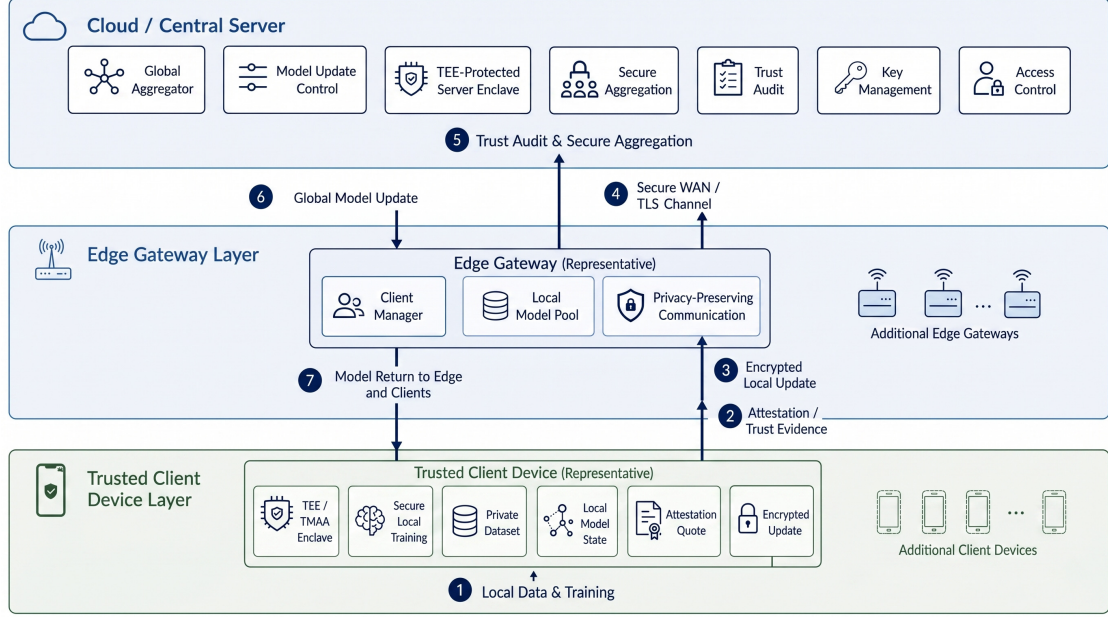


Figure 1: Federated computing network interaction: TEE-anchored client-edge-cloud architecture

2.4 Trusted Hardware Root of Trust in Federated Environments (TEE)

To fill the gap in physical trust, research has begun to introduce trusted execution environments (TEEs) such as Intel SGX and ARM TrustZone. Through isolated execution and protected memory, TEEs preserve the integrity of critical code and key material, making it difficult for an attacker to directly tamper with the attestation process even after the host operating system has been compromised[3, 25]. Existing work shows that TEEs have practical value in training integrity[4], threat mitigation[26], and trust sharing in industrial IoT[27]. Their federated extensions and efficiency optimization in SGX environments have also been validated[28, 29]. On this basis, this paper builds a hardware-software collaborative defense system. Unlike methods that only perform static encryption or single-dimensional anomaly evaluation, this work temporally couples hardware remote attestation with dual-stream reputation evolution. The cloud side is responsible for parameter auditing, while the edge-side TMAA provides trusted evidence and runtime monitoring, forming a coordinated protection mechanism that combines mandatory admission with dynamic circuit breaking.

3 System Model and Problem Definition

3.1 Client-Edge-Cloud Collaborative Architecture and System Workflow Modeling

This paper considers a typical client-edge-cloud heterogeneous communication environment (see Figure 1). The system consists of a central **aggregation server (Server, $\mathcal{S}$)** and K edge clients, denoted by $\mathcal{C} = \{c_1, c_2, \dots, c_K\}$. Clients may join or leave dynamically, and some nodes may be controlled by attackers.

Based on the network interaction topology above, the proposed trust-flow-driven trusted federated framework builds a complete trust transmission and evaluation chain between the edge and the cloud. The overall workflow of the system and the relationships among its main physical entity modules are defined as follows:

1. **Edge client side (trusted execution and feature extraction):** Each admitted client must be equipped with a trusted execution environment (TEE) and a trusted management

agent (TMAA) deployed inside it. During local training, the TMAA generates an Attestation Quote before training to statically verify the integrity of the execution environment. It also continuously monitors key behavioral feature sequences during training, such as changes in gradient direction and loss fluctuation. Information entropy and Kalman filtering are then used for an initial dynamic trust assessment, which is uploaded to the cloud together with the model update parameters.

2. **Central server side (auditing, evolution, and aggregation control):** As the global coordinator, the server receives client updates and trust evidence, then performs a three-stage coordinated defense based on trust flow. It first conducts **content auditing**, extracting a clean reference direction and using cosine similarity to calculate high-dimensional update quality. It then enters **dual-stream state evolution**, where Beta Bayesian inference is used to build a historical utility stream for long-term contribution evaluation, while multidimensional max-value probes update an instant risk stream to identify sudden anomalies. Finally, **layered robust aggregation** applies differentiated gated clipping and survivor weight re-normalization according to the security sensitivity of different network layers, producing and distributing a screened global update.

Through the coordinated interaction of the edge and cloud entity modules, the system forms a full-lifecycle physical and logical closed loop: hardware admission control → runtime feature monitoring → trust state evolution → risk-gated aggregation.

3.2 Composite Threat Assumptions and Strong Trust-Base Boundary (Threat Model)

To clearly define the boundary between attacker and defender capabilities, this paper adopts the following threat model (see Figure 2):

- **Explicit TEE boundary:** Intel and AMD TEE-capable platforms protect attestation-critical code, device keys, report integrity, report freshness, and the trusted monitoring path against a compromised host OS. They do not certify labels, samples, optimization objectives, or semantic intent. Thus, a client can produce a signed poisoned update inside an otherwise valid measured training workflow; post-attestation modification outside that workflow is excluded.
- **Honest-but-curious server:** The server follows the dual-stream and layered-aggregation protocol and may inspect individual updates and audit statistics, but it cannot change the stated transition rules.
- **Internal malicious clients:** An admitted client may perform label flipping, trigger backdoors, clean-label poisoning, semantic poisoning, sign flipping, or gradient scaling. The analytical assumption is fewer than 50% malicious admitted clients; the 50% experiment is an out-of-assumption stress test, not a Byzantine threshold or a theoretical guarantee. Physical probing and advanced microarchitectural side channels are excluded.

3.3 Design Goals and System Boundary

The design goals of the proposed framework are to maintain convergence under the stated attacks, reduce irreversible false bans of long-tail clients, and report overhead by system boundary: the appended TrustReport is about 15 KB, server aggregation plus audit increases from 2.10 s to 4.85 s, and the auxiliary secure path measures approximately 11 ms per admitted-client round for enclave transitions, memory pressure, and cryptographic aggregation. Separately, device-side TEE Quote generation, signing, and lightweight monitoring are measured at approximately 12–15 ms

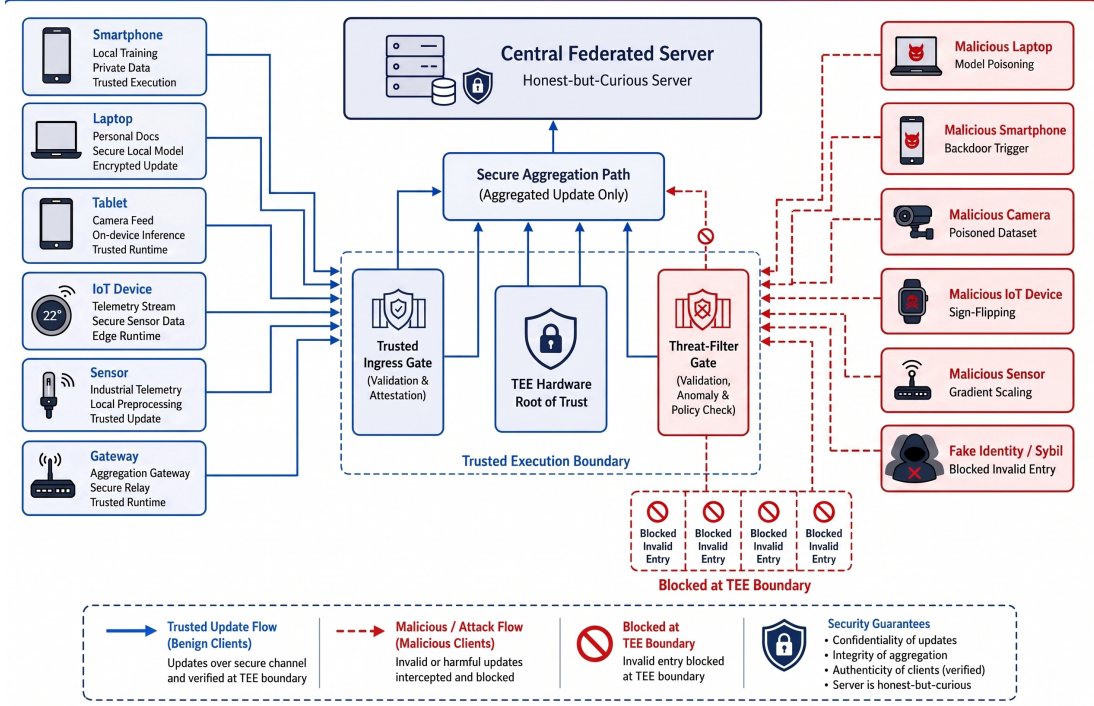


Figure 2: Security penetration and multidimensional composite poisoning threat model in an edge wide-area network

per admitted-client round. Full end-to-end communication and vendor-specific attestation latency remain outside the reported measurements. The term secure aggregation is not used here: the server must inspect per-client evidence, so cryptographic privacy-preserving aggregation is outside the system boundary.

4 Proposed Method: Trust-Flow Framework

For the threat model above, this paper proposes a trust-flow-driven trusted federated learning framework (see Figure 3). Following the execution order of federated learning, the framework first performs trusted admission and runtime monitoring on the client side, then conducts current-round content auditing and cross-round reputation evolution on the server side, and finally enters layered security control during aggregation. Each training round is therefore divided into four decision stages: admission and runtime monitoring, content auditing, state evolution, and layered aggregation; global model distribution follows as the protocol output. At a higher level of abstraction, these stages correspond to three layers: client admission and monitoring, server auditing and evolution, and aggregation security assurance.

The key of this macro-to-micro structure is the continuous transfer of trust states. Phase one outputs *TrustScore*, indicating whether the client has a trustworthy basis for admission and execution. Phase two outputs *ContentScore*, judging whether the update submitted in the current round deviates from the clean reference and the collaborative direction of the group. Phase three deposits current-round signals into *HistPerf* and *RiskEMA*, distinguishing long-term low contribution from short-term high risk. Phase four fuses the three types of signals into *RawScore* and performs differentiated aggregation by network layer. Unlike single-point defenses, data generated at each phase continues to constrain later phases, continuously updating the **Trust Flow** and forming closed-loop control.

To facilitate the following derivation, Table 1 lists the definitions of core symbols.

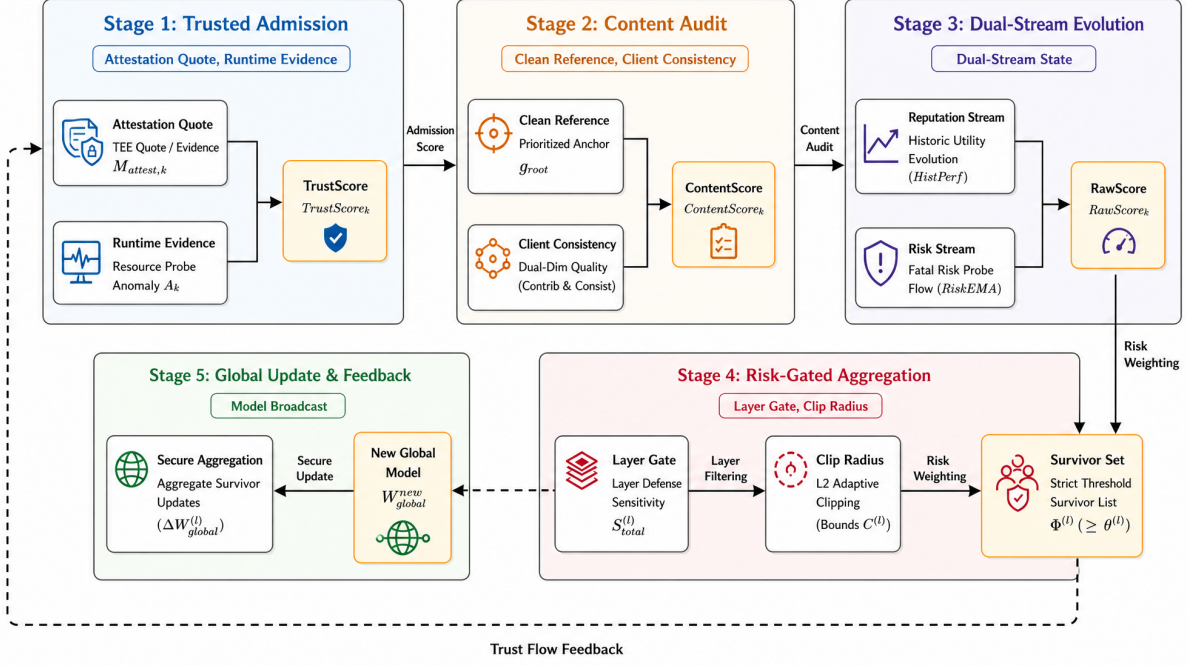


Figure 3: Overview of the four-stage defense loop in trust-flow-driven trusted federated learning

4.1 Hardware Admission and Dynamic Trust Assessment

Admission is a two-stage process: static execution admission and dynamic evidence tracking. Let $M_{attest,k} \in \{0, 1\}$ indicate whether the attestation report, freshness nonce, and binding to the current round and update digest are valid. During initialization, the client invokes a device-unique key and produces a quote containing a vendor-specific measurement digest, code identity, runtime configuration, round nonce, update digest, and a monotonic freshness counter. A client that fails this check is rejected before its update enters statistical aggregation.

During local training, the trusted management and attestation agent monitors the standardized sequence $\mathcal{X}_k^{(t)} = \{\Delta grad, \Delta loss, instr_dist\}$. After concatenating the standardized feature histograms, let $h_k^{(t)} = H(\mathcal{X}_k^{(t)})/H_{max} \in [0, 1]$. We define the entropy-derived observation variance and observation as

$$R_k^{(t)} = \exp(\lambda h_k^{(t)}), \quad Z_k^{(t)} = \text{clip}\left(w_a M_{attest,k} + (1 - w_a)(1 - h_k^{(t)}), 0, 1\right), \quad (3)$$

where $w_a = 0.7$ and $\lambda = 1.0$. High entropy represents uncertain evidence; it is not used as a direct maliciousness label.

The scalar Kalman recursion is

$$T_k^{-,(t)} = T_k^{(t-1)}, \quad P_k^{-,(t)} = P_k^{(t-1)} + Q, \quad (4)$$

$$\begin{aligned} K_k^{(t)} &= \frac{P_k^{-,(t)}}{P_k^{-,(t)} + R_k^{(t)}}, \\ T_k^{(t)} &= T_k^{-,(t)} + K_k^{(t)} \left(Z_k^{(t)} - T_k^{-,(t)} \right), \\ P_k^{(t)} &= (1 - K_k^{(t)}) P_k^{-,(t)}. \end{aligned} \quad (5)$$

Here $Q = 0.01$ is the fixed process-noise variance. A fresh admitted client is initialized with

Table 1: Core system parameters and trust-flow notation

Symbol	Definition	Symbol	Definition
$W_{global}^{(t)}$	Global model weights in round t	$\mathcal{A}$	Active node set eligible for valid aggregation in the current round
ΔW_k	Model gradient/parameter update submitted by client k	$\Phi^{(l)}$	Valid surviving subset at layer l
$TrustScore_k$	TEE hardware-anchored admission trust score	$HistPerf_k$	Accumulated state of the historical utility stream ($HistPerf \in [0, 1]$)
$ContentScore_k$	Current-round content quality score under dual references	$RiskEMA_k$	Decayed state of the instant risk stream ($RiskEMA \in [0, 1]$)
$RawScore_k$	Final privilege score after fusing three-dimensional scores and applying risk discounting	$M_{attest,k}$	Hardware remote attestation verification flag
g_{root}	Clean guiding anchor direction extracted by the server	A_k	Runtime anomaly score of client k in physical resource usage

$T_k^{(0)} = 1$, $P_k^{(0)} = 0.25$, and $RiskEMA_k^{(0)} = 0$. The admission privilege is

$$TrustScore_k^{(t)} = M_{attest,k} \text{clip}\left(\frac{T_k^{(t)}}{1 + P_k^{(t)}}, 0, 1\right). \quad (6)$$

High entropy therefore lowers the Kalman gain and prevents an uncertain observation from abruptly rewriting cross-round trust. A new client starts in NORMAL after valid attestation; its first-round evidence is provisional and cannot trigger an irreversible blacklist.

4.2 Reference Direction Construction and Consistency Auditing

Relying only on cosine similarity among clients can be misled by collusion attacks. The server therefore prioritizes a declared clean reference when available. In the reported experiments, a balanced 500-sample proxy set (50 images per CIFAR-10 class) is held by the server, is not included in client training, and is provided to every reference-dependent baseline under the same protocol. The same set supports the guiding gradient, clean-loss probe, and activation probe. If this direction is unavailable, a fallback reference is constructed from high-trust, low-risk admitted clients. The corresponding weight is defined as:

$$\omega_k^{ref} = TrustScore_k \cdot (1 - RiskEMA_k^{prev})^2. \quad (7)$$

The final reference direction g_{root} is:

$$g_{root} = \begin{cases} g_{root_clean}, & \text{if server validation direction available,} \\ \sum_k \frac{\omega_k^{ref}}{\sum_j \omega_j^{ref}} \Delta W_k, & \text{otherwise.} \end{cases} \quad (8)$$

After obtaining g_{root} , the server calculates a content quality score for each client update ΔW_k . This score considers both global contribution, namely consistency with the reference direction, and group consistency, namely the degree of agreement with other high-reputation clients:

$$S_{contrib,k} = \max(0, \alpha_1 + \alpha_2 \cdot \cos(\Delta W_k, g_{root})), \quad (9)$$

$$S_{consist,k} = \frac{\sum_{j \neq k} TrustScore_j \cdot \xi_{j,k}}{\sum_{j \neq k} TrustScore_j + \varepsilon}, \quad (10)$$

$$ContentScore_k = \frac{(1 + \beta_{fusion}^2) \cdot S_{consist,k} \cdot S_{contrib,k}}{\beta_{fusion}^2 \cdot S_{consist,k} + S_{contrib,k} + \varepsilon}. \quad (11)$$

Here, $\xi_{j,k} = \max(0, \alpha_1 + \alpha_2 \cos(\Delta W_k, \Delta W_j))$, and $\alpha_1 + \alpha_2 = 1.0$ adjusts the cosine mapping interval. The parameter $\beta_{fusion} > 1.0$, with default $\beta_{fusion} = 2.0$, moderately increases the weight assigned to consistency with the clean direction, strengthening resistance to collusive drift. $ContentScore_k$ describes only the content quality of the current-round update and is not equivalent to permanent reputation. For legitimate long-tail nodes under strong Non-IID settings, a low single-round score may result from data skew rather than attack behavior. This score is therefore passed to the cross-round state evolution in phase three, where subsequent handling is jointly determined by the historical utility stream and the instant risk stream.

Algorithm 1 Admission and content auditing based on TEE hardware anchors and clean reference

Input: $\mathcal{C}$, $\{\Delta W_k\}_{k \in \mathcal{C}}$, $RiskEMA^{(t-1)}$

Output: $TrustScore_k$, $ContentScore_k$, valid reference baseline g_{root}

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1: Phase 1: Hardware Anchoring and Dynamic Trust Measurement
2: for each client  $k \in \mathcal{C}$  do
3:   if  $M_{attest,k} == 0$  (TEE remote attestation fails) then
4:     Reject admission:  $TrustScore_k \leftarrow 0$ , and skip this round
5:   else
6:     Inside TEE: record behavior sequence  $\mathcal{X}_k = \{\Delta grad, \Delta loss, instr\_dist\}$ 
7:     Compute information entropy  $H_k \leftarrow -\sum p(x_i) \log p(x_i)$ 
8:     Map entropy to measurement noise  $R_k \leftarrow \exp(\lambda \cdot H_k)$ 
9:     Kalman filter update:
10:    Trust estimate  $\hat{T}_k$ , covariance  $P_k \leftarrow \text{KalmanFilter}(H_k, \text{previous state})$ 
11:    Admission trust score  $TrustScore_k \leftarrow \hat{T}_k \cdot \frac{1}{1+P_k}$ 
12:   end if
13: end for
14: Phase 2: Reference Direction Construction and Collaborative Consistency Auditing
15: if server clean validation data are available then
16:   Extract true guiding gradient  $g_{root} \leftarrow g_{root\_clean}$ 
17: else
18:   Compute nonlinear weight  $\omega_k^{ref} \leftarrow TrustScore_k \cdot (1 - RiskEMA_k^{(t-1)})^2$ 
19:   Generate fallback clean anchor  $g_{root} \leftarrow \sum_k (\omega_k^{ref} \Delta W_k) / \sum_j \omega_j^{ref}$ 
20: end if
21: for each admitted active client  $k$  do
22:   Global contribution score  $S_{contrib,k} \leftarrow \max(0, \alpha_1 + \alpha_2 \cos(\Delta W_k, g_{root}))$ 
23:   Group collaboration score  $S_{consist,k} \leftarrow \frac{\sum_{j \neq k} TrustScore_j \max(0, \alpha_1 + \alpha_2 \cos(\Delta W_k, \Delta W_j))}{\sum_{j \neq k} TrustScore_j + \varepsilon}$ 
24:   Harmonic content score  $ContentScore_k \leftarrow \frac{(1 + \beta_{fusion}^2) S_{consist,k} S_{contrib,k}}{\beta_{fusion}^2 S_{consist,k} + S_{contrib,k} + \varepsilon}$ 
25: end for
26: return  $\{TrustScore_k, ContentScore_k, g_{root}\}$ 

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4.3 Dual-Stream Evolution of Historical Utility and Instant Risk

A single reputation score confuses low utility caused by data skew with high risk caused by an attack. Trust Flow therefore maintains a slow historical utility stream and a fast current-risk stream. Let $q_k^{(t)} = \text{clip}(\text{ContentScore}_k^{(t)}, 0, 1)$. The historical stream uses the decayed Beta update

$$a_k^{(t)} = \beta_h a_k^{(t-1)} + q_k^{(t)}, \quad b_k^{(t)} = \beta_h b_k^{(t-1)} + 1 - q_k^{(t)}, \quad \text{HistPerf}_k^{(t)} = \frac{a_k^{(t)}}{a_k^{(t)} + b_k^{(t)} + \varepsilon}. \quad (12)$$

A fresh client starts with $a_k^{(0)} = b_k^{(0)} = 1$, so $\text{HistPerf}_k^{(0)} = 0.5$. The decay factor $\beta_h \in (0, 1)$ is fixed before the test rounds. The historical stream reduces privilege gradually and allows a legitimate long-tail client to recover.

The short-term risk stream is

$$\text{RiskEMA}_k^{(t)} = \beta_r \text{RiskEMA}_k^{(t-1)} + (1 - \beta_r) \text{Risk}_k^{\text{inst}}, \quad (13)$$

where $\text{Risk}_k^{\text{inst}}$ is the maximum current probe. A fresh, validly attested client starts in NORMAL. Let $u_k^{(t)} = 1 - \text{HistPerf}_k^{(t)}$ and $r_k^{(t)} = \text{RiskEMA}_k^{(t)}$. The thresholds $(\tau_h, \tau_s, \tau_q, \tau_b) = (0.26, 0.64, 0.80, 0.90)$ govern utility deficit, soft risk, quarantine, and blacklisting. A NORMAL client enters SUSPECT after two consecutive rounds with $u_k^{(t)} > \tau_h$ or $r_k^{(t)} \geq \tau_s$. A SUSPECT client enters QUARANTINE after three such consecutive rounds or one current probe with $\text{Risk}_k^{\text{inst}} \geq \tau_q$. A QUARANTINE client enters BLACKLIST after four consecutive rounds with $r_k^{(t)} \geq \tau_b$. Recovery from SUSPECT or QUARANTINE requires two consecutive rounds with $u_k^{(t)} \leq \tau_h$ and $r_k^{(t)} < \tau_s$; BLACKLIST is irreversible within an experiment. Counters reset when their conditions are not met, and the state update is completed before layer aggregation. First-round evidence can reduce the current update privilege but cannot trigger an irreversible blacklist.

The streams meet only in the aggregation privilege:

$$\text{RawScore}_k^{(t)} = (\text{TrustScore}_k^{(t)})^\alpha (\text{ContentScore}_k^{(t)})^\beta (\text{HistPerf}_k^{(t)})^\gamma (1 - \text{RiskEMA}_k^{(t)})^p. \quad (14)$$

This timing allows current risk to veto a sleeper burst despite high history, while a low-similarity benign client remains recoverable through the slow stream.

4.4 Risk-Gated Layered Aggregation

Backdoors may concentrate in deep layers, while shallow feature layers can carry useful heterogeneous information. Trust Flow therefore assigns each layer l a sensitivity score. With L total layers, define

$$S_{\text{priv}}^{(l)} = \exp(-\tau_p l / L), \quad S_{\text{util}}^{(l)} = \frac{\|g_{\text{root}}^{(l)}\|_2}{\max_m \|g_{\text{root}}^{(m)}\|_2 + \varepsilon}, \quad (15)$$

$$S_{\text{sec}}^{(l)} = \frac{1}{2} \left(1 - \frac{1}{|\mathcal{T}|} \sum_{k \in \mathcal{T}} \cos(\Delta W_k^{(l)}, g_{\text{root}}^{(l)}) \right). \quad (16)$$

Here $\mathcal{T} = \{k \in \mathcal{A} : M_{\text{attest},k} = 1, \text{RiskEMA}_k^{(t-1)} < \tau_s\}$ is the trusted reference set. If it is empty, the server uses $\mathcal{A}$ only for normalization and records the absence of a trusted peer reference. Thus $S_{\text{priv}}^{(l)}$, $1 - S_{\text{util}}^{(l)}$, and $S_{\text{sec}}^{(l)}$ lie in $[0, 1]$. We set

$$S_{\text{total}}^{(l)} = w_p S_{\text{priv}}^{(l)} + w_u (1 - S_{\text{util}}^{(l)}) + w_s S_{\text{sec}}^{(l)}, \quad (w_p, w_u, w_s) = (0.3, 0.3, 0.4). \quad (17)$$

The layer threshold and clipping radius are

$$\theta^{(l)} = \mu_{\text{base}} + \lambda_s S_{\text{total}}^{(l)}, \quad C^{(l)} = \frac{C_{\text{base}}}{S_{\text{total}}^{(l)} + \varepsilon_c}, \quad (18)$$

where $\tau_p = 3.0$ and C_{base} is the median trusted-client update norm. A layer with stronger suspicious drift is therefore harder to enter and more tightly clipped.

Survivors are $\Phi^{(l)} = \{k \in \mathcal{A} : RawScore_k^{(t)} \geq \theta^{(l)}\}$. To preserve the stated global objective, data-volume weights are retained:

$$\tilde{w}_k^{(l)} = \frac{p_k RawScore_k^{(t)}}{\sum_{j \in \Phi^{(l)}} p_j RawScore_j^{(t)} + \varepsilon}, \quad \widehat{\Delta W}_k^{(l)} = \frac{\Delta W_k^{(l)}}{\max(1, \|\Delta W_k^{(l)}\|_2 / C^{(l)})}. \quad (19)$$

The layer update is $\Delta W_{global}^{(l)} = \sum_{k \in \Phi^{(l)}} \tilde{w}_k^{(l)} \widehat{\Delta W}_k^{(l)}$. If no client survives, the client with maximum *RawScore* is retained with unit normalized weight and the same clipping radius. Re-normalization prevents the removal of risky clients from silently reducing the effective learning rate.

4.5 Model Update Distribution and Knowledge Retention

Upon the completion of layered aggregation, the server applies the screened increment to the global model, $W_{global}^{new} = W_{global}^{old} + \Delta W_{global}$, and distributes the updated model together with the retained client states, including *HistPerf*. This closes the control loop for the stated protocol; convergence and attack suppression remain empirical properties evaluated in the experiments.

4.6 Interpretation of the Dual-Stream Design

The historical stream acts as a low-pass filter for benign content variation, whereas the maximum probe and the risk EMA preserve abrupt warnings. Under bounded benign fluctuations and persistent attack evidence, this separation explains why a long-tail client can recover while a sleeper burst is rapidly down-weighted. These observations are design properties under the stated thresholds; they are not a theorem guaranteeing zero FPR or universal Byzantine robustness. Formal guarantees would require distributional assumptions and attack models beyond the present empirical scope.

5 Experiments and Analysis

5.1 Experimental Setup and Reproducible Parameters

5.1.1 Baseline Environment and Hardware-Software Stack

The experiments use Flower 1.5.0 and PyTorch on CIFAR-10 with a CIFAR-10-adapted ResNet-18. The first convolution uses a 3×3 kernel with stride one, the initial max-pooling layer is removed, and the classifier has ten outputs. Training and trusted execution use two real hardware configurations based on Intel and AMD TEE-capable processors. NVIDIA A10 GPUs execute model training, while CPU-side TEEs isolate attestation, trusted monitoring, report generation, and admission-state processing. Both platforms execute the same request/response protocol, nonce-based freshness verification, measured-code identity checking, round and update-digest binding, protected report generation, and admission decision. The testbed consists of anonymized dual-CPU servers and five NVIDIA A10 GPUs, running Ubuntu 20.04.6 LTS and CUDA 12.8.

Five independent seeds are run on each platform. Aggregate means and standard deviations are computed over the resulting ten platform-seed runs, while state trajectories use one fixed seed per platform for readability. The reported runtime measures server-side aggregation and auditing; it excludes client communication and vendor-specific attestation latency. Enclave transitions, memory pressure, and the cryptographic aggregation path are measured separately at approximately 11 ms per admitted-client round. Device-side TEE Quote generation, signing, and lightweight monitoring are measured separately at approximately 12–15 ms per admitted-client round.

5.1.2 Dataset Heterogeneous Partitioning

The 50,000 CIFAR-10 training images are partitioned into a shared 5,000-image client pool, replicated as a common base subset for all clients, and 45,000 exclusive images: clients 0–9 use an IID split, clients 10–14 use a Dirichlet partition with $\alpha = 1.0$, and clients 15–19 use a strong long-tail partition with $\alpha = 0.1$. Separately, the server holds a balanced 500-image proxy set (50 per class) from CIFAR-10 test data; it is not used in client training and is provided to every reference-dependent baseline. FedAvg, Krum, and Trimmed Mean do not consume the proxy set. This separation makes the clean-data assumption explicit and distinguishes the shared client-training pool from the server-side audit reference.

5.1.3 Malicious Attack Instances and Proportions

The formal 30% setting injects six malicious clients into 20 admitted clients. The six behaviors are active concurrently in every local training round. Client 0 performs untargeted label flipping with $\gamma = 0.5$. Client 1 injects a fixed 3×3 lower-right trigger into 20% of local samples and targets class 0. Client 2 applies label-preserving clean-label perturbations to 20% of local samples with the same target. Client 3 applies a fixed semantic attribute bias to 20% of local samples and assigns the biased subset to class 0. Client 4 performs untargeted sign flipping, and Client 5 scales its update by a factor of 100. Clients 1–3 are targeted attacks; Clients 0, 4, and 5 are evaluated through clean loss, convergence, and state-transition metrics. Five Sybil nodes without valid certificates and three free-rider nodes with forged low workload are used only to validate Phase-One gating and are intercepted before statistical aggregation. Malicious updates that pass attestation represent an attacker operating inside an admissible training workflow; post-attestation host tampering is excluded by the threat model. Each run uses 30 rounds and three local epochs.

For each targeted behavior $a \in \{\text{trigger}, \text{clean}, \text{semantic}\}$, let $\mathcal{B}_a$ be its trigger- or attribute-bearing test subset and y_a^* its target. We define $ASR_a = |\{x \in \mathcal{B}_a : \arg \max f(x) = y_a^*\}|/|\mathcal{B}_a|$ and report $ASR_{\text{target}} = \frac{1}{3} \sum_a ASR_a$. The three attack-specific values are retained in the convergence traces. Untargeted label, sign, and scaling attacks are evaluated separately through clean accuracy, loss, and state-transition time.

Preemptive Interception Clarification: The aforementioned eight external infiltration or resource fraud nodes are uniformly intercepted during the protocol phase. They either fail the TMAA hardware gating ($M_{\text{attest},k} = 0$) or see their $\text{TrustScore}_k \rightarrow 0$ owing to anomalous behaviors. Consequently, the 20 nodes statistically analyzed in subsequent sections represent the official participants that successfully navigated the Phase One admission.

The global hyperparameters and defensive component configurations employed in the experiment are delineated in Table 2.

Execution Details: Each run executes 30 rounds of federated training, with 3 local epochs per round. Aggregate accuracy and ASR use the ten platform-seed runs; trajectory charts use one fixed seed per platform. No interpolation smoothing is applied. The reported duration is a server-side aggregation and audit measurement rather than an end-to-end deployment time.

5.2 Empirical Analysis of Defense Effectiveness

5.2.1 Global Convergence and Attack Prevention Performance

As illustrated in Figure 4, under a 30% malicious proportion, the model accuracy converges steadily to 92.31%, while the backdoor ASR plummets to 10.21%.

Statistical Reliability Clarification: This section and subsequent aggregate metrics use the mean and standard deviation over ten platform-seed runs. The node trajectory charts originate from one fixed seed per platform.

50% Malicious Proportion Stress Test: Because the analytical assumption is fewer than 50% malicious admitted clients, an auxiliary *Flwr-half* environment with 10 malicious and 10

Table 2: Global Hyperparameters and Defense Component Configuration

Module	Hyperparameter Description	Value
Base Environment	Total nodes in heterogeneous topology (K)	20
	Local SGD parameters (lr , $Momentum$)	0.001, 0.9
Admission and State	Admission weight w_a , entropy scale λ , process noise Q	0.7, 1.0, 0.01
	Initialization ($T_0, P_0, RiskEMA_0$)	(1, 0.25, 0)
	2D audit mapping α_1, α_2 and β_{fusion}	0.6, 0.4, 2.0
	History smoothing β_h and risk half-life β_r	0.9, 0.85
	State thresholds ($\tau_h, \tau_s, \tau_q, \tau_b$)	0.26, 0.64, 0.80, 0.90
	Persistence for SUSPECT/QUARANTINE/BLACKLIST	2, 3, 4 rounds
Layer Gating Phase 4	Score fusion exponents (α, β, γ, p)	3.0, 1.0, 0.5, 1.0
	Threshold lower bound μ_{base} and penalty scale λ_s	0.4, 1.2
	Sensitivity weights (τ_p, w_p, w_u, w_s)	(3.0, 0.3, 0.3, 0.4)

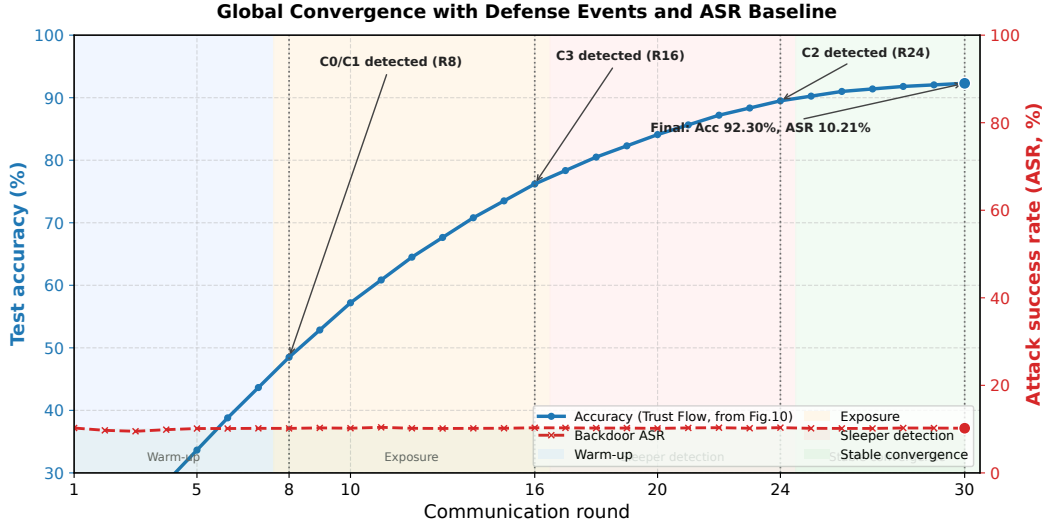


Figure 4: Global Convergence Accuracy and Backdoor ASR Evolution Curves

legitimate nodes is reported only as an out-of-assumption stress test. It is not used to infer a theoretical boundary. The outcomes are detailed in Table 3.

In this stress test, the framework reaches 91.81% accuracy and 10.46% targeted ASR, while all 10 malicious nodes are removed under the final-ban metric and no legitimate node is permanently banned in the tested run. The 30-round loss trajectory is a descriptive observation of this configuration; it does not constitute a guarantee outside the stated threat model.

5.2.2 Trust Flow Tracing and Malicious Node Banning Analysis

Figure 5 shows representative state trajectories for explicit poisoning, trigger and semantic backdoors, parameter attacks, and one benign long-tail client. The attack clients accumulate risk through the direction, amplitude, clean-loss, activation, or peer probes and are eventually quarantined or blacklisted according to the persistence rules. The long-tail client may enter SUSPECT or QUARANTINE after a low content score, but it recovers when the risk stream does not persist. These trajectories illustrate the state machine under one fixed seed; they are not universal detection-time claims.

Table 3: Extreme 50% Malicious Ratio Stress Test vs. Standard 30% Baseline

Setting	Acc (%)	ASR (%)	TPR (%)	FPR (%)
Standard Baseline (30% malicious, 6/20)	92.31 $\pm$ 0.09	10.21 $\pm$ 0.05	100	0.00
Stress Test (50% malicious, 10/20)	91.81 $\pm$0.10	10.46 $\pm$0.06	100	0.00

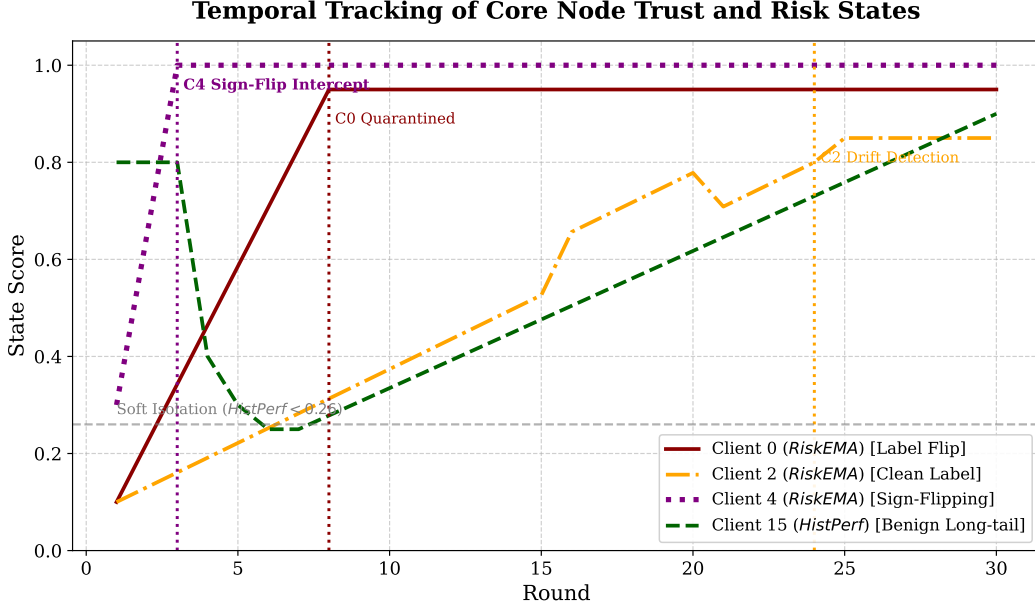


Figure 5: Temporal Tracking of Core Node Trust and Risk States under Mixed Attacks

5.2.3 False Positive Protection for Long-Tail Nodes (FPR Analysis)

For Trust Flow, zero legitimate nodes trigger a permanent ban across the reported 30-round runs. This final-ban FPR is a method-specific state metric; stateless baselines are marked N/A rather than assigned an artificial blacklist state. Even if certain legitimate nodes (e.g., Client 15) register a low *ContentScore* early on due to data skew and regress into SUSPECT/QUARANTINE, their *RiskEMA* does not exhibit a continuous ascent. Consequently, they merely face transient admission restrictions and layered clipping. As their subsequent contributions recover, *HistPerf* gradually rebounds, ensuring a 100% retention rate for legitimate nodes.

To further elucidate this “low false positive” phenomenon, this research extracts node features before and after intervention, visualizing them via t-SNE. Figure 6 exhibits that the dual-stream mechanism effectively disentangles the malicious clusters from the legitimate long-tail clusters, simultaneously anchoring the legitimate heterogeneous nodes within the principal aggregation domain.

5.3 Lateral Baseline Comparison of Defense Mechanisms

Fairness Statement: All baselines are executed under the identical federated configuration as this paper. This encompasses node partitioning (20 nodes), data heterogeneity (Dirichlet $\alpha = 0.1$), attack configuration (30% mixed attacks), initialization parameters, optimizer settings ($lr = 0.001$, $momentum = 0.9$), and communication rounds.

Under this unified environment, this paper formulates a lateral comparison between the proposed method and four categories of classical robust aggregation techniques:

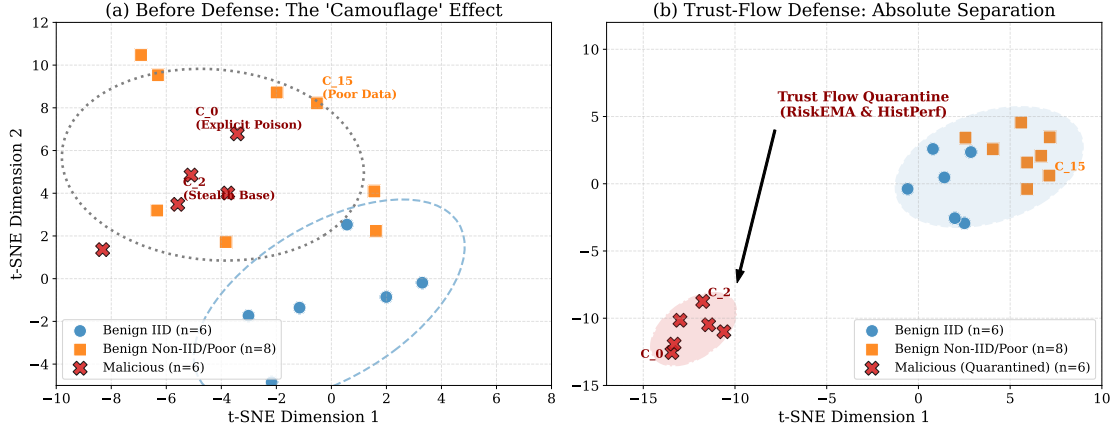


Figure 6: t-SNE Visualization of Node Feature Vectors Before and After Dual-Stream Defense

Table 4: Performance Comparison of FL Defense Strategies under 30% Mixed Attacks

Defense Strategy	Core Mechanism	Global Acc. $\mu \pm \sigma$ (Acc. %)	ASR $\mu \pm \sigma$ (ASR %)	Benign FPR μ (FPR %)
FedAvg [32]	No defense baseline	86.15 $\pm$ 1.2	92.34 $\pm$ 5.1	N/A (*)
Krum [9]	Euclidean distance	64.20 $\pm$ 2.8	12.15 $\pm$ 0.9	N/A (*)
Trimmed Mean [10]	Coordinate-wise trimming	71.35 $\pm$ 2.4	38.60 $\pm$ 4.2	N/A (*)
FLTrust [7]	Unidirectional trust anchor	81.25 $\pm$ 1.5	15.42 $\pm$ 1.1	N/A (*)
Proposed (Trust Flow)	Dual-stream + layered clipping	92.31 $\pm$ 0.09	10.21 $\pm$ 0.05	0.00

(*) Note: FedAvg, Krum, Trimmed Mean, and FLTrust do not implement the Trust Flow permanent blacklist state machine. Their FPR is therefore not comparable under this definition and is marked as N/A.

FedAvg[32] serves as the defenseless baseline, mirroring the degree of model degradation when subjected to attacks; **Krum**[9] selects a singular representative update via minimal Euclidean distance; **Trimmed Mean**[10] executes truncated averaging across coordinate dimensions; **FLTrust**[7] leverages a small-scale clean set on the server to construct a reference vector, applying weighting based on cosine similarity.

The quantitative outcomes of each method in this high-risk scenario are documented in Table 4.

Within the matched classical comparison, FedAvg reaches 86.15% accuracy with 92.34% targeted ASR, while Krum, Trimmed Mean, and FLTrust trade accuracy for lower ASR. Trust Flow reaches 92.31% accuracy and 10.21% macro-averaged targeted ASR under the same protocol. These values do not establish a universal ranking: recent defenses such as DeepSight, FLAME, CrowdGuard, RoseAgg, FLPurifier, ShieldFL, and RaSA require interfaces that are not reproduced here, so their omission is treated as a comparison limitation. No statistical-significance claim is made beyond the reported repeated-run summary.

5.4 Component-Level Ablation Study

Addressing the common "black-box" performance evaluations in current federated security research, this section replaces coarse-grained phase superimposition tests with component ablation experiments. This methodology examines how core modules, including dual-stream orthogonal decoupling, multi-dimensional risk probes, and layer-wise dynamic re-normalization, independently

Table 5: Ablation Analysis of Orthogonal Decoupling on FPR and Recall

Reputation Architecture	Threshold Control	FPR	TPR
Single-Stream EMA (Baseline)	Aggressive (anti-poison)	18.25%	95.12%
	Conservative (anti-FP)	2.10%	48.33%
HistPerf-Only	Dynamic adaptive	0.15%	21.05%
Dual-Stream Decoupling (Proposed)	Dynamic adaptive	0.00%	100.00%

contribute to defensive robustness and system availability.

5.4.1 Effectiveness of Dual-Stream Reputation Decoupling

To substantiate that the system’s exceptionally low false positive rate (FPR) stems not from conservative defensive thresholds, but intrinsically from the orthogonal decoupling of historical utility and instantaneous risk, two degraded variants are constructed for comparison: a Single-Stream Exponential Moving Average (Single-Stream EMA) and a HistPerf-Only model. Table 5 delineates the precise interception performance of these divergent mechanisms under the intense pressure of severe Non-IID conditions and a 30% mixed attack scenario.

Experimental data reveals a clear trade-off in the single-stream mechanism when it must accommodate heterogeneous updates and malicious poisoning: tightening the threshold misclassifies 18.25% of innocent long-tail nodes, whereas relaxing the threshold reduces TPR against sleeper attacks. Relying exclusively on the historical utility stream accommodates long-tail features (FPR 0.15%), but its interception rate against sleeper-burst attacks is only 21.05% because it lacks sensitivity to short-term anomalies. The dual-stream mechanism separates long-term contribution accumulation from short-term risk circuit breaking, removing all malicious clients under the final-ban metric while avoiding permanent false bans on benign clients. The FPR/TPR values in Table 5 adopt the final-ban definition. As a supplementary statistic, when the recoverable SUSPECT state is counted as a risk-review signal, its per-round coverage over malicious clients is 88.33%, and its recoverable review trigger rate over benign clients is 14.79%; this state is used only for auditing and down-weighting, and does not constitute a permanent false ban.

5.4.2 Multi-Dimensional Risk Probe Ablation

When verifying the efficacy against defensive penetration, a solitary reliance on a clean reference set frequently falls short of comprehensively neutralizing multi-dimensional compound attacks. Figure 7 contrasts the ASR suppression capabilities of merely employing a pristine reference (Vanilla Clean-Root), superimposing shallow statistical probes (+ Shallow Probes), and deploying full-dimensional deep probes (+ Full Probes) when confronted with untargeted flipping, gradient scaling, and stealthy Clean-Label Backdoor attacks.

The visualization clarifies the complementary role of each probe. For Sign-Flipping, whose direction anomaly is explicit, the clean reference already limits ASR to 12.5%, and the shallow gradient probe further reduces it to 8.1%. For Gradient Scaling, where abnormal update magnitude is more salient, the shallow probe reduces ASR from 24.3% to 11.2%. Clean-Label Backdoor is more difficult: its malicious update remains highly collinear with benign directions, so the ASR is 42.1% with only the clean reference and remains 28.5% after adding shallow probes. After enabling deep probes based on cross-entropy (r_{probe}) and high-level feature KL divergence ($r_{trigger}$), the system captures fine-grained functional and representation shifts caused by hidden triggers and reduces the attack success rate to 9.2%.

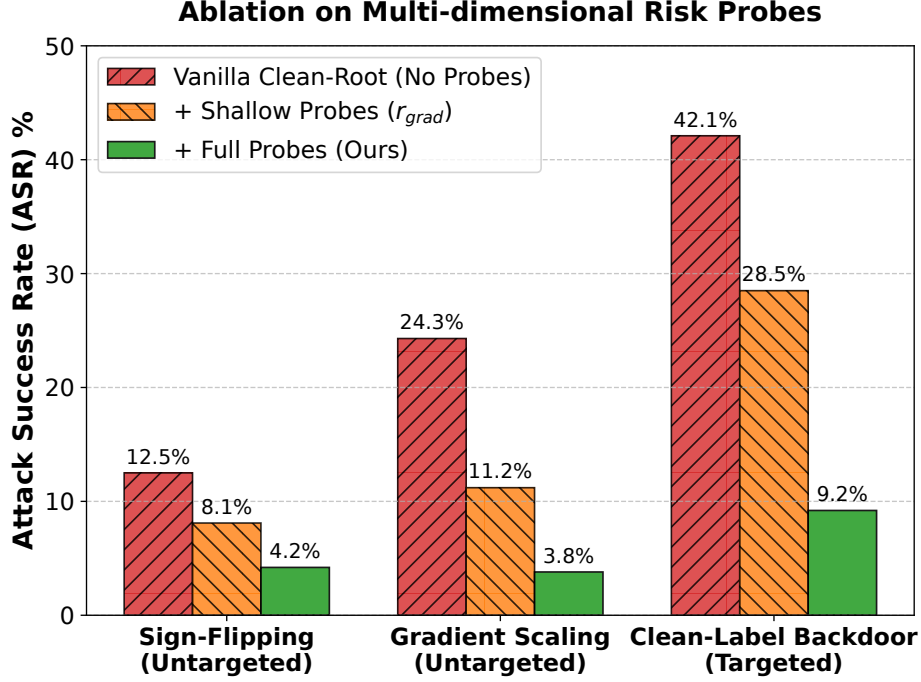


Figure 7: ASR Suppression Comparison of Multi-Dimensional Risk Probes across Attack Types

5.4.3 Layered Gating and Re-Normalization Convergence Benefits

Subsequent to the successful ablation of attacks, securing the rapid restoration of the global model’s availability emerges as another pivotal concern. Figure 8 evaluates the disparities in convergence velocity and terminal accuracy among a traditional, indiscriminate global clipping mechanism (Global-Clipping), the execution of layer-wise gating bereft of weight reallocation (Hierarchical w/o Renorm), and the holistic framework introduced herein (Trust-Flow Proposed).

The trajectory trends starkly indicate that the aggressive global L2 clipping profoundly dilutes the model’s generalization capabilities by excessively interfering with the weight updates of the shallow feature extraction layers, effectively stalling the ultimate accuracy at approximately 85%. Although incorporating layer-wise gating ameliorates this deficiency to an extent (elevating accuracy to 89%), the elimination of malicious nodes precipitates a scenario where the effective aggregation weights of the current round fall short of the unit constraint (i.e., $\sum \tilde{w} < 1$). This instigates an “implicit learning rate decay,” thereby ensuring the convergence curve remains retarded. The constrained projection and weight re-normalization mechanisms introduced in this study dynamically compensate for the influence of surviving nodes. This strategy enables stable convergence within the 30-round training budget and restores the model accuracy to a level close to the attack-free setting (92.4%), fully preserving system utility while maintaining uncompromising security.

5.5 Sensitivity Analysis on Data Heterogeneity

A plethora of robust aggregation methods lean heavily upon IID conditions. To evaluate the stability of the proposed framework under a same-family distribution shift, the global accuracy was tested across weak-to-strong Dirichlet concentration settings while keeping the label space, model, optimizer, and attack protocol fixed. The outcomes are depicted in Figure 9.

The curves show that Krum and FLTrust are competitive under weak heterogeneity but degrade as the concentration decreases toward the strong long-tail regime. Trust Flow is less sensitive in the reported sweep because current content is fused with historical utility and short-term risk. This is

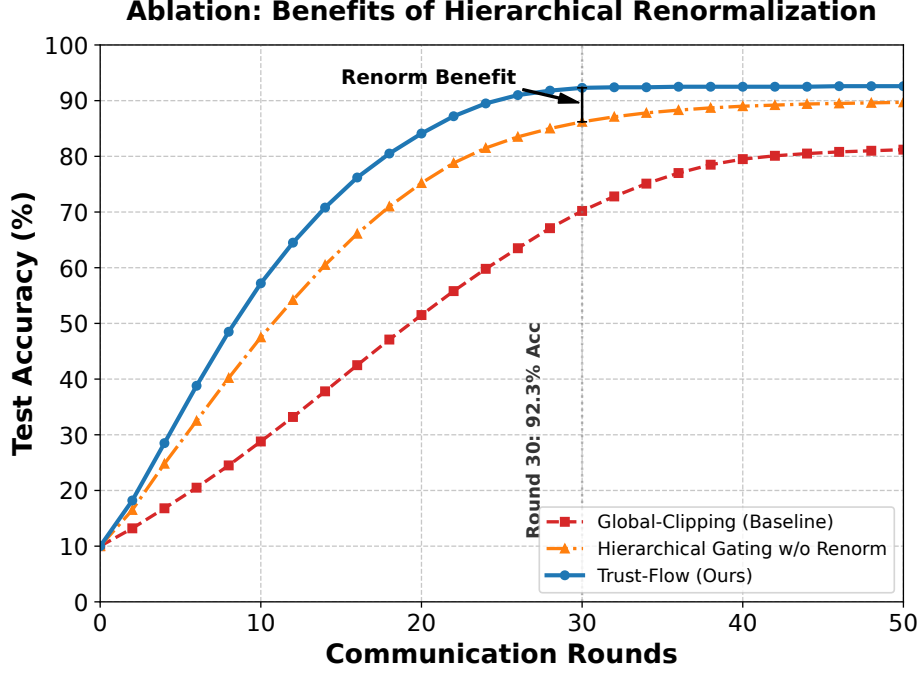


Figure 8: Convergence and Accuracy Benefits of Dynamic Weight Re-Normalization

Table 6: Edge-Cloud Overhead: Standard FL vs. Proposed Framework (20 Clients)

System Architecture	Edge Extra Latency	Upload Extra Payload	Cloud Aggregation Time
Standard FedAvg	0 ms (Baseline)	0 KB (Baseline)	~ 2.10 s
Trusted FL (Proposed)	12–15 ms; 11 ms	~ 15 KB (appended report)	~ 4.85 s (server audit only)

an empirical robustness observation under the tested family of partitions, not a distribution-free guarantee.

Furthermore, discrete testing at 0.5, 1.0, 2.0, and 3.0 was conducted on the pivotal parameter β_{fusion} . A smaller magnitude fosters the retention of long-tail nodes but simultaneously escalates ASR risks; conversely, an oversized value skews the aggregation excessively towards homogeneous updates. Weighing the trade-offs between Acc and ASR, $\beta_{fusion} = 2.0$ was definitively selected.

5.6 System Computation and Communication Overhead Analysis

The overhead measurement separates the server-side trust-flow cost from end-to-end deployment cost. The TrustReport adds approximately 15 KB per transmission. The reported 2.10 s to 4.85 s increase covers server-side aggregation and audit only; it excludes communication and vendor-specific attestation latency. The auxiliary secure path measures approximately 11 ms per admitted-client round for enclave transitions, memory pressure, and the cryptographic aggregation path. This auxiliary path is reported separately from the server-side result because the main protocol still requires per-client evidence inspection. Separately, device-side TEE Quote generation, signing, and lightweight monitoring add a measured 12–15 ms per admitted-client round.

Layer-wise cosine evaluation increases server computation. In the empirical trials, server aggregation and audit time increased from 2.10 s to 4.85 s. This measurement reports server-side protocol cost; end-to-end deployment acceptability still depends on communication and vendor-

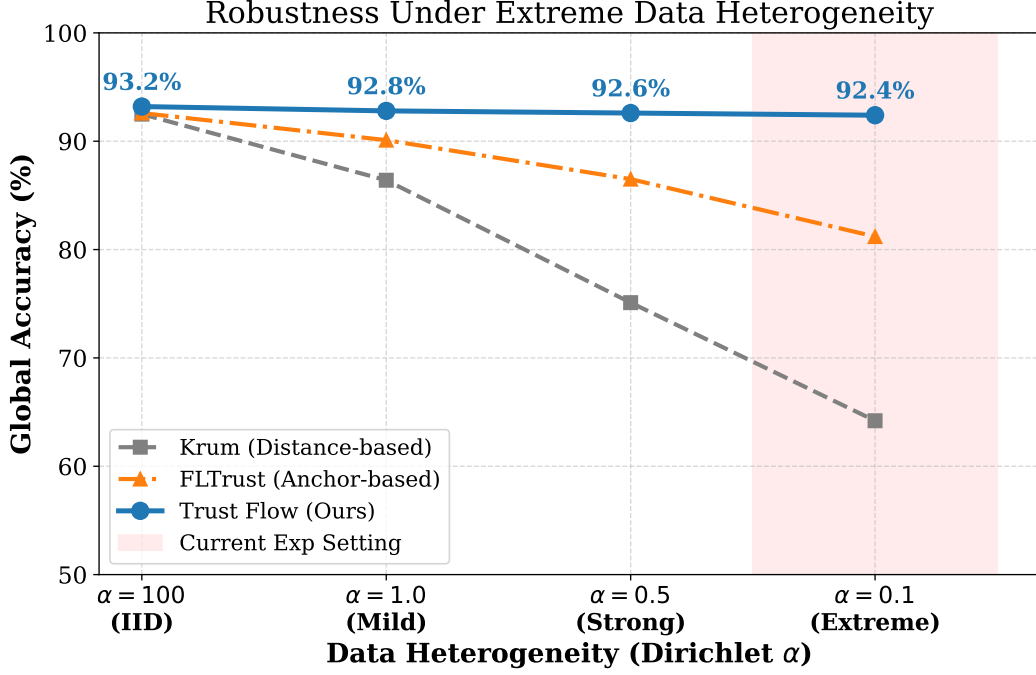


Figure 9: Robustness Stress Test under Varying Data Heterogeneity (Dirichlet α)

specific attestation characteristics that were not measured here.

6 Conclusion and Future Work

This paper presents Trust Flow as a system-level pipeline that orders and separates execution admission, current content auditing, temporal state evolution, and layer-wise aggregation. Under the stated 30% concurrent-attack protocol, it reaches 92.31% clean accuracy and 10.21% macro-averaged targeted ASR, with no permanent false ban in the tested runs. The 50% experiment is an out-of-assumption stress test and does not establish a theoretical Byzantine boundary.

Limitations and future work: The clean reference and activation probes require an explicitly declared server-held proxy set; the peer-reference branch is defined but not evaluated in the headline results. The comparison uses matched classical baselines and does not directly reimplement every recent temporal, backdoor, or privacy-preserving defense. The probe design is tailored to vision classification, and full end-to-end attestation and communication latency, including vendor-specific components, were not measured; device-side TEE Quote generation, signing, and lightweight monitoring were measured separately at approximately 12–15 ms per admitted-client round, while enclave transitions, memory pressure, and the cryptographic aggregation path were measured separately as an auxiliary secure path. Future work should evaluate reference-free settings, matched modern baselines, larger deployments, and non-vision federated tasks.

A Detailed Computation of Instant Risk Flow Probes

To augment reproducibility and transparency, this appendix mathematically delineates the calculation details of the core probes (r_{grad} , r_{probe} , $r_{trigger}$) underpinning the instant risk stream originally introduced in Section 4. During the t -th aggregation round, the probes are defined as follows:

A.1 A.1 Gradient Direction and Amplitude Probe (r_{grad})

The r_{grad} probe quantifies the geometric deviation of the client update $\Delta W_k^{(t)}$ relative to the reference direction $g_{root}^{(t)}$, while r_{amp} captures an amplitude anomaly relative to the admitted clients. Both quantities are clipped to $[0, 1]$:

$$r_{grad,k}^{(t)} = \text{clip}\left(\frac{1 - \cos(\Delta W_k, g_{root})}{2}, 0, 1\right), \quad (20)$$

$$r_{amp,k}^{(t)} = \text{clip}\left(\frac{\|\Delta W_k\|_2}{\text{median}_j \|\Delta W_j\|_2 + \varepsilon} - 1, 0, 1\right). \quad (21)$$

where $\text{clip}(x, 0, 1) = \min(1, \max(0, x))$ and ε avoids division by zero.

A.2 A.2 Small-Sample Cross-Entropy Probe (r_{probe})

Relying exclusively on distance statistics is grossly inadequate for identifying targeted pollution directed against minority classes. The r_{probe} leverages a small-scale, pristine probe set on the server, denoted as $\mathcal{D}_{probe}$, calculating the incremental surge in cross-entropy loss (Loss Surge) pre- and post-merger:

$$r_{probe,k}^{(t)} = \text{clip}\left(\frac{\mathcal{L}_{probe}(W + \Delta W_k) - \mathcal{L}_{probe}(W)}{\delta_{clean} + \varepsilon}, 0, 1\right). \quad (22)$$

Here δ_{clean} is the loss-increment scale calibrated from trusted reference updates.

A.3 A.3 Deep Neuron Backdoor Activation Probe ($r_{trigger}$)

Countering profoundly concealed threats like Clean-Label attacks, standard statistical and validation losses often fall short. The $r_{trigger}$ discerns anomalies through the distributional variance of high-level feature activations. Assuming A_k denotes the client model’s pre-activation outputs on the validation set, and $\mathcal{H}(\cdot)$ signifies the Softmax normalization into a probability simplex to satisfy the KL-Divergence prerequisites:

$$r_{trigger,k}^{(t)} = \text{clip}\left(\frac{\text{KL}(\mathcal{H}(A_{base}) \parallel \mathcal{H}(A_k))}{\delta_{trigger} + \varepsilon}, 0, 1\right). \quad (23)$$

Here $\delta_{trigger}$ is the calibrated activation-drift scale from trusted reference updates.

For completeness, the peer channel is defined by the trust-weighted cosine disagreement

$$\bar{c}_k^{(t)} = \frac{\sum_{j \neq k} \text{TrustScore}_j c_{kj}}{\sum_{j \neq k} \text{TrustScore}_j + \varepsilon}, \quad r_{peer,k}^{(t)} = \text{clip}(1 - \bar{c}_k^{(t)}, 0, 1), \quad (24)$$

where $c_{kj} = \cos(\Delta W_k, \Delta W_j)$. After normalization, the channels are fused by a maximum so that a strong warning is not averaged away:

$$Risk_k^{inst} = \max\{r_{grad,k}, r_{amp,k}, r_{probe,k}, r_{trigger,k}, r_{peer,k}\}. \quad (25)$$

This single-round risk value enters the EMA pipeline for the subsequent circuit-breaking decision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Jiakai Zhou: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. Chao Bu: Supervision, Project administration, Funding acquisition, Writing – review & editing.

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Data availability

The data and materials that support the findings of this study are available from the corresponding author upon reasonable request.

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